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Spatial, mobility, or socio-economic inequity? A district level job accessibility evaluation in Jakarta, Indonesia

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ABSTRACT

Jakarta, a rapidly urbanizing megacity in Southeast Asia, faces persistent challenges in ensuring equitable access to employment opportunities. This study evaluates spatial, socio-economic, and modal inequities in job accessibility by examining how disparities manifest across three transport modes – car, motorcycle, and public transport – and different income groups. Using a cumulative opportunity approach within a 41.2-minute travel threshold, the analysis draws on travel time data collected from the Google Maps API and origin-destination matrices derived from a microsimulation model. Results show that job accessibility is highly centralized, with significantly lower access in peripheral and low-density areas. Motorcycles provide the broadest and most equitably distributed access, especially for low- and medium-income populations, while public transport shows the most severe inequality. The Theil index reveals that job accessibility inequity in Jakarta is driven primarily by disparities within population density groups, highlighting the need for localized, neighbourhood-level interventions rather than broad spatial or income-based approaches. Notably, Gini coefficients across income groups are nearly identical, suggesting that transport inequity in Jakarta is shaped more by geography and network design than by socio-economic category alone. These findings highlight the need for expanding mass public transport coverage, improving first- and last-mile connectivity, and implementing localized interventions in underserved neighbourhoods. The study contributes empirical evidence to the literature on urban accessibility and equity in the Global South and supports the integration of multi-dimensional equity metrics into transport planning practices.

ARTICLE HISTORY

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KEYWORDS

Job accessibility; equity; Gini coefficient; mobility; Palma ratio; Theil index

Highlights

- District-level analysis of job accessibility inequity in Jakarta, Indonesia.
- Spatial centrality drives unequal job access, disadvantaging peripheral areas.

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- Motorcycle mobility delivers most equitable access across income groups.
- Public transport access is highly unequal, concentrated in limited corridors.
- Inequity shaped more by geography and mode than by socio-economic class.

1. Introduction

The increasing rate of urbanization has brought significant mobility challenges, particularly concerning how urban populations travel, the available transportation opportunities, and its accessibility level. Hansen (1959) initially defined accessibility as the potential for opportunities to interact. Subsequently, accessibility has been defined in a variety of ways. For instance, Dalvi and Martin (1976) defined accessibility as the convenience with which any activity (land use) can be accessed using the existing transportation system. Access to activities is the primary goal of transportation, influenced by factors such as individual mobility, the quality of available transportation modes, transportation network connectivity, as well as population density and land use variation (Litman, 2013).

Among the diverse indicators of accessibility, job accessibility emerges as a central focus due to its critical role in socio-economic outcomes. Jobs are a fundamental driver of economic mobility and social inclusion. Access to employment opportunities directly affects individuals' economic well-being, making job accessibility a key indicator for understanding spatial and socio-economic inequalities (Guzman et al., 2017; Lucas, 2012). Although many studies on transportation equity evaluate access to healthcare or education (e.g. Lardier et al., 2023; Sharma & Patil, 2021; Tiznado-Aitken et al., 2021), job accessibility is frequently used due to its direct impact on economic outcomes and daily commuting needs (Bastiaanssen et al., 2021; Lucas et al., 2016; Martens, 2016). Unlike healthcare, which may be needed less frequently, jobs require consistent and reliable access, making employment-related trips a critical factor in transportation planning.

Numerous studies have demonstrated the uneven distribution of job accessibility in many large cities worldwide. For instance, Dodson et al. (2011) in several Australian cities, Teunissen et al. (2015) in Bogota, Hernandez (2018) in Montevideo, Kawabata (2003) in Tokyo, and (Kawabata & Shen, 2007) in the US metropolitan areas. This unequal job accessibility and its association with limited opportunities in certain areas and social interaction exacerbate pre-existing socio-economic inequalities (Blanco & Apaolaza, 2018; Golub & Martens, 2014). Identifying the most affected or marginalized groups and areas is crucial to evaluating the role of transportation systems in promoting equity (Lucas et al., 2016). Recent transportation studies have shown interest in measuring differences in job accessibility and its impact on socially marginalized groups (see for example, Boisjoly et al., 2017; Grisé et al., 2019; Liu & Kwan, 2020).

Despite this growing body of work, limited research focuses on the Global South – particularly Southeast Asia – where transport systems are characterized by mixed modal patterns and the dominance of informal and motorcycle-based mobility. Moreover, few studies have applied multidimensional equity indicators to capture both modal and socio-economic disparities in job accessibility. In Asian countries, such as Indonesia, evaluations of job accessibility are not widely conducted or able to explore how

accessibility is distributed across socio-economic strata and transport modes. The study by Andani et al. (2021) on accessibility to employment in Indonesia in the context of evaluating toll road construction, for example, does not accommodate the aspect of mode choice. In fact, the characteristics of mobility in Asian countries are very unique and different from countries on other continents. For example, the use of (semi) automatic motorbikes with low engine capacity (below 150 cc) is very widespread in Indonesia and other developing countries such as Vietnam and Thailand (Hsu et al., 2003). This is in contrast to motorbike users in Europe and the United States. Ride-hailing services with motorbikes are also more popular in Southeast Asia, and have been found to improve mobility in urban communities (Irawan et al., 2020; Phun et al., 2019; Wadud, 2020). Ride-hailing services provide a more flexible alternative for low-income communities that rely heavily on public transport (Qiao & Gar-On Yeh, 2023).

The government of Indonesia has prioritized the development of mass public transport in the past years as part of their strategy to boost mobility and accessibility. As of the end of 2019, types of mass public transport such as Bus Rapid Transit, MRT, and LRT were operational in Jakarta. It is said that transportation services in addition to what existed before were able to service about 90 percent of the people. However, there has been no analysis of how the expansion of these services has affected access to employment or reduced socio-economic inequities in mobility. This study seeks to address that gap by evaluating inequities in job accessibility across socio-economic groups and transport modes in Jakarta.

The unequal distribution of resources or opportunities among population groups or places, such as public transport accessibility between central and periphery districts, is called inequality in transport (Arellana et al., 2021). Inequity emphasizes fairness and justice beyond inequality (Litman, 2015). Systemic limitations to transportation services can prevent disadvantaged populations from participating in critical activities like job, education, and healthcare. While inequality can be quantified by statistical measures like the Gini coefficient, inequity necessitates assessing whether observed discrepancies are fair or unfair based on ethical and socio-economic factors (van Wee & Geurs, 2011). This study investigates inequities in job accessibility to identify gaps and offer recommendations that promote fairness in mobility and opportunity distribution across districts and social groups.

This research provides critical empirical evidence from Jakarta, a highly relevant case in the Global South, using equity metrics rarely applied in the region. It examines job accessibility inequity through the combined lens of spatial, socio-economic, and modal disparities in a city where motorcycles dominate mobility and public transport systems are still emerging. While previous studies have focused on transit affordability or infrastructure distribution in other cities, this paper uses the Gini coefficient, Palma ratio, and Theil index to evaluate job accessibility from a multidimensional perspective. By doing so, the study provides one of the first empirical analyses of accessibility inequity in Indonesia that considers the interplay of income levels, geographic location, and mode of transport. It responds to recent calls in the literature for more nuanced and context-sensitive assessments of transport-related inequality (Lucas, 2019; Pereira et al., 2016; Uteng & Lucas, 2017).

This paper is organized as follows: Section 2 presents a literature review focused on transportation equity and job accessibility, highlighting essential metrics employed in the assessment of inequities. Section 3 outlines the research methodology, including data collection, travel time analysis, and the computation of accessibility and inequity indicators. Section 4 presents the results and includes a comprehensive discussion that

compares the findings with existing literature. Section 5 concludes the paper by summarizing key insights and providing policy recommendations aimed at enhancing equitable job access in Jakarta.

2. Literature review

To examine inequity in urban transport systems, this study is guided by a conceptual framework that integrates three key dimensions of transport-related disparities: spatial inequity, socio-economic inequity, and mobility inequity (Figure 1). These dimensions represent distinct but interconnected pathways through which access to employment is shaped by geographic location, income status, and modal availability. All three influence individuals' ability to reach job opportunities, conceptualized in this study as job accessibility. This framework also links each dimension of inequity with suitable distributional equity indicators: the Gini coefficient and Lorenz curve for overall inequality, the Palma ratio for disparities between extremes, and the Theil index for analysing within- and between-group differences. This multi-metric approach enables a nuanced evaluation of transport-related exclusion in Jakarta. Each component of the framework is discussed in detail in the following subsections.

2.1. Understanding transport inequity and its dimensions

Transport inequity refers to the unfair or unjust distribution of transport services, infrastructure, and the opportunities they enable. It differs from inequality, which is

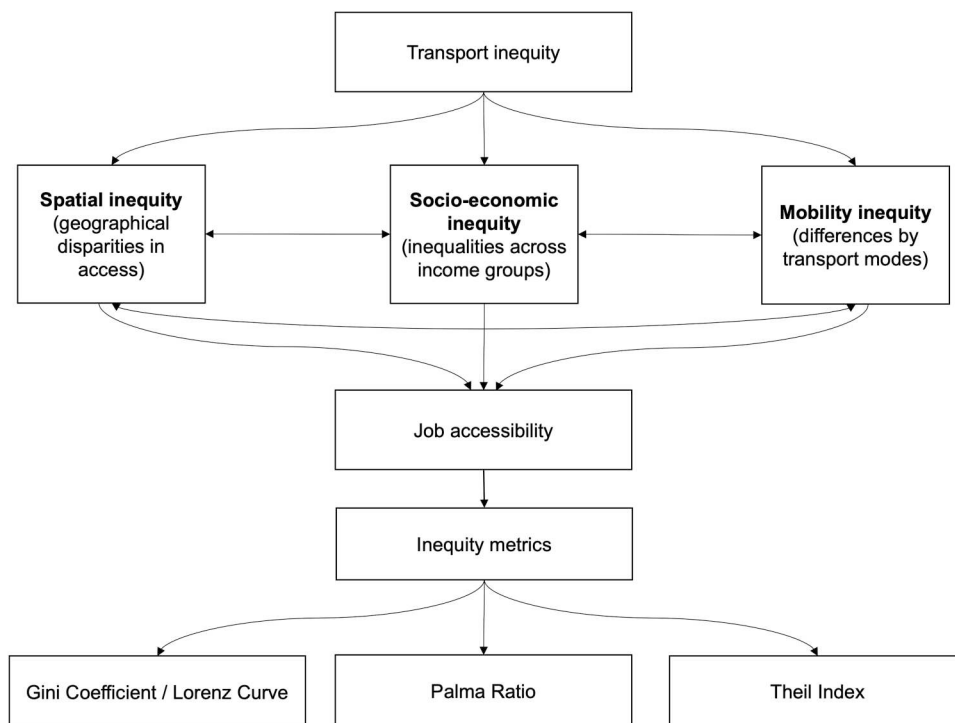


Figure 1. Conceptual framework.

descriptive in nature, by introducing a normative evaluation of whether the observed disparities are ethically or socially acceptable (Litman, 2015; van Wee & Geurs, 2011). Transport equity concerns whether individuals and communities – regardless of income, location, or demographic characteristics – can access opportunities and participate fully in society. This study focuses on three key and interrelated dimensions of transport inequity, which are spatial, socio-economic, and mobility inequities. The terms spatial inequity, socio-economic inequity, and mobility inequity address disparities in opportunities and resources, each emphasizing distinct dimensions.

Spatial inequity arises from geographic disparities in the distribution of transport infrastructure and services. In many cities, central or high-income areas benefit from dense networks and high service levels, while peripheral or low-density zones often face limited connectivity and longer travel times (Arellana et al., 2021; Panzera & Postiglione, 2020). Spatial inequity is assessed through the analysis of the distribution of resources, services, and opportunities across various geographical regions. Indicators of significance encompass access to public services, including healthcare and education, income inequality, unemployment rates, housing quality, and employment opportunities (Lawal & Anyiam, 2019; Morasae et al., 2012).

Socio-economic inequity concerns disparities in access based on income, education, or employment status. Lower-income groups may face financial barriers, longer commutes, and reduced modal choice, which constrain their access to urban opportunities (Lucas, 2012). Socioeconomic inequity is assessed by analysing disparities in social and economic status across different demographic groups, including factors such as income, education, and occupation. Common measures encompass income distribution, the percentage of the population residing below the poverty line, and variations in employment rates or educational attainment (Hertel & Groh-Samberg, 2019).

Mobility inequity entails two primary domains: socioeconomic mobility and geographic or physical mobility. Socioeconomic mobility examines the ease with which individuals can enhance their social and economic status throughout their lives or across generations, shaped by elements such as educational access, job quality, and inheritances (Chetty et al., 2014; Corak, 2013). In contrast, geographic or physical mobility refers to the capacity of individuals to navigate within a city or region, influenced by factors such as the availability of transport options (Brockmann et al., 2006). Limited access to reliable public transit or car ownership can restrict geographic and physical mobility, which in turn limits individuals' access to job opportunities, education, and essential services, thereby impeding socioeconomic mobility. Assessment is conducted based on factors including the availability, affordability, and quality of transport, average commuting times, vehicle ownership rates, and the accessibility of destinations through walking, cycling, or public transport. This study examines mobility inequity as a type of transportation-related disparity.

Spatial inequity, socioeconomic inequity, and physical mobility inequity are inter-related concepts that influence one another in intricate manners. Spatial inequity influences socio-economic inequity by determining the distribution of resources and opportunities among various social groups, often resulting in the concentration of wealth and services in particular regions while leaving others underserved (Gezici & Hewings, 2007; J. J. Shen et al., 2022). Spatial inequity restricts physical mobility, as individuals in regions with inadequate transport access encounter difficulties in accessing

employment opportunities or essential services, thereby perpetuating socio-economic disparities (Petersen et al., 2010).

Socioeconomic inequity significantly influences the disparities in spatial and physical mobility. Individuals from lower socioeconomic backgrounds often experience restricted access to reliable transport, which impedes their mobility within urban or regional areas (Tomasello et al., 2022). The limitation of physical mobility can exacerbate socio-economic disparities by hindering access to education, healthcare, and employment opportunities, thereby obstructing socio-economic mobility (Morasae et al., 2012).

In addition, disparities in physical mobility can exacerbate spatial and socioeconomic inequalities. Disparities in access to transport infrastructure can result in spatial divides, leading to the marginalization of specific areas due to inadequate connectivity (Sun et al., 2020). The absence of physical mobility options can isolate communities from economic opportunities and essential services, thereby perpetuating socio-economic disparities and spatial inequalities. These three forms of inequity are interconnected and function in a cyclical manner. Spatial inequity restricts physical mobility and intensifies socio-economic inequity, whereas socio-economic inequity can hinder physical mobility and sustain spatial inequity.

2.2. Accessibility and job accessibility as analytical lenses

Accessibility is a key concept in transport planning and equity analysis. It refers to the ease with which people can reach desired destinations, such as jobs, education, healthcare, or other essential services (Dalvi & Martin, 1976; Hansen, 1959). Unlike mobility – which focuses on the capacity to move – accessibility highlights the relationship between transport systems, land use, and the spatial distribution of opportunities (Litman, 2013). In equity evaluations, accessibility offers a more outcome-oriented perspective, as it reflects people's ability to participate in socio-economic life.

Among different types of accessibility, job accessibility is especially significant due to its direct implications for economic welfare and inclusion. Employment access is a daily necessity, and the quality of this access can influence income levels, time poverty, and long-term socio-economic mobility (El-Geneidy et al., 2016; Lucas et al., 2016). While many transport equity studies have examined access to healthcare (e.g. Jin et al., 2022; Sharma & Patil, 2021; J. J. Shen et al., 2022) or education (e.g. Pizzol et al., 2021; Tiznado-Aitken et al., 2021), job accessibility is more sensitive to daily transport conditions and often exposes deep-rooted spatial and socio-economic divides. For example, Boisjoly et al. (2017) examined job accessibility in São Paulo, Brazil, found that access is predominantly concentrated in high-income areas, primarily attributable to employment informality and the unequal distribution of public transport services.

Similar other studies also indicates that low-income populations in big cities in Australia, Brazil, Uruguay, Great Britain, the United States, and many more face extended commutes, elevated transport costs, and restricted access to public transit, which limits their employment opportunities (see, for example, El-Geneidy et al., 2016; Fan et al., 2012; Guzman et al., 2017; Kawabata & Shen, 2007; Sharma & Patil, 2021; Q. Shen, 2001; Smith et al., 2020; Teunissen et al., 2015). These studies address spatial mismatch and modal inequity, while also highlight the significance of affordable transport and high-quality network infrastructure.

In cities of the Global South – particularly in Southeast Asia – there is a relative scarcity of research that examines job accessibility in a comprehensive, multi-dimensional way. Urban mobility in these contexts is shaped by unique modal mixes, including motorcycles and informal transport services, and rapidly evolving infrastructure systems. This study addresses these gaps by assessing job accessibility across income groups and transport modes in Jakarta, where motorcycles and ride-hailing services are widely used and public transport remains unevenly distributed.

2.3. Operationalizing accessibility-based inequity: metrics and indicators

Accessibility measures are crucial for analysing transportation inequity as they reveal the varying effects of transportation systems on different socio-economic groups (Golub & Martens, 2014). Nonetheless, they possess certain limitations. Metrics like the Gini coefficient and Lorenz curve (Gini, 1936) offer general evaluations of inequality; however, they may fail to reflect nuanced disparities among particular subgroups. These measures do not consider individuals' socioeconomic status, making it impossible to determine whether inequalities disproportionately affect disadvantaged populations. They also lack directionality, meaning they cannot distinguish between progressive and regressive changes in accessibility. To address this issue, various metrics such as the Theil index (Conceição & Ferreira, 2000) and Palma ratio (Palma, 2011) can be integrated to offer a more comprehensive understanding of inequities related to accessibility.

The Gini coefficient is used to assess overall inequality in the distribution of job accessibility. It provides a single summary statistic that reflects the concentration of access among the population, where higher values indicate greater inequity (Delbosc & Currie, 2011). The Palma ratio assesses the accessibility distribution between the top 10% and the bottom 40% of the population, emphasizing disparities at the extremes (Guzman & Oviedo, 2018). The Theil index decomposes overall inequity into within-group and between-group components, facilitating an analysis of inequity variation within and between categories, such as population density groups (Jin et al., 2022).

These indicators translate abstract inequity concepts into observable and comparable patterns across transport modes, zones, and population groups. By applying them to job accessibility outcomes – disaggregated by income group, transport mode, and spatial zone – the study offers a structured and quantifiable assessment of transport-related exclusion. This operational framework builds on global best practices while adapting to the specific urban and modal characteristics of a Southeast Asian megacity.

3. Data and methods

The analytical framework of this study integrates spatial and temporal travel data with job distribution metrics to evaluate job accessibility and its inequity in Jakarta. As illustrated in Figure 2, the methodology comprises two main stages: accessibility estimation and inequity measurement. In the first stage, accessibility is calculated using the cumulative opportunities method within a 41.2-minute travel time threshold (Jakarta's average commuting time – JICA, 2019). This involves combining travel time data – parametrized by mode and collected via Google Maps API on November 2, 2020 at 10:00 AM – with job distribution data derived from origin-destination (OD) matrices. The second stage

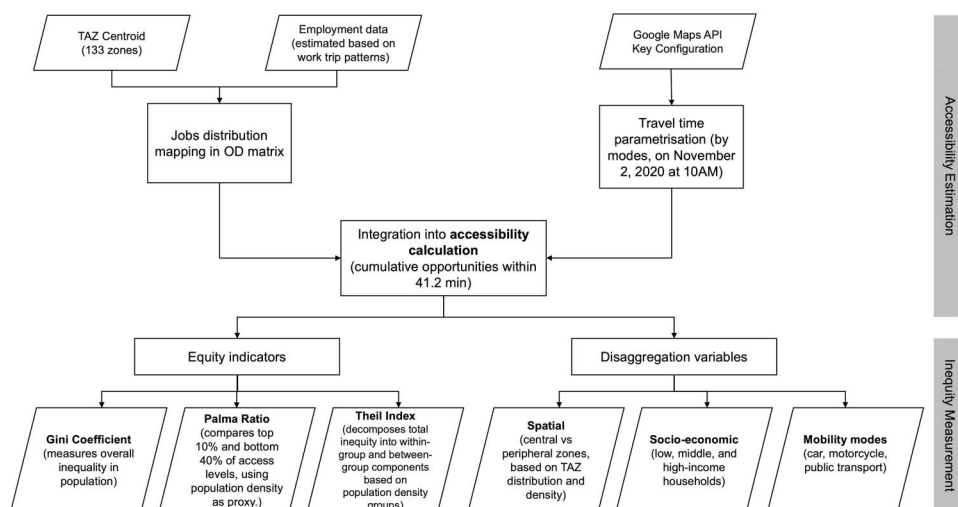


Figure 2. Analytical framework.

applies equity indicators (Gini coefficient, Palma ratio, and Theil index) to assess the distribution of accessibility across disaggregated variables, including transport modes, income levels, and population density. This integrated approach enables a nuanced evaluation of spatial, socio-economic, and modal inequities in job accessibility across Jakarta. More detailed descriptions of the data sources, accessibility calculation, and equity assessment techniques are provided in the following subsections.

3.1. Study area and traffic analysis zone (TAZ)

Jakarta was selected as the case study for this research due to its complex urban morphology, rapidly growing population, and persistent challenges related to transport inequity. As Indonesia's capital and largest city, Jakarta presents a highly relevant context for investigating accessibility disparities, particularly in light of its evolving transport landscape, which includes recent investments in mass public transport such as the MRT, LRT, and BRT (Abiad et al., 2019; Hasibuan & Mulyani, 2022; Martinez & Masron, 2020). The city is also characterized by a high degree of modal heterogeneity, where formal systems coexist with informal and motorcycle-based travel. These characteristics make Jakarta a representative and policy-relevant case for understanding how geography, infrastructure distribution, and socio-economic diversity shape job accessibility outcomes in a Global South megacity.

Jakarta Province, comprising five administrative cities: North Jakarta, South Jakarta, East Jakarta, West Jakarta, and Central Jakarta. The study area (Figure 3) is segmented into Traffic Analysis Zones (TAZs), facilitating a comprehensive evaluation of travel behaviour throughout the region. The definition of TAZs follows the methodology established by the Japan International Cooperation Agency (JICA, 2019).

It indicates that the distribution of framework data for formulating traffic analysis zones utilizes a combination of multi-regression analysis and a GIS-based 'development potential model', incorporating various statistical and spatial data. These methods enable

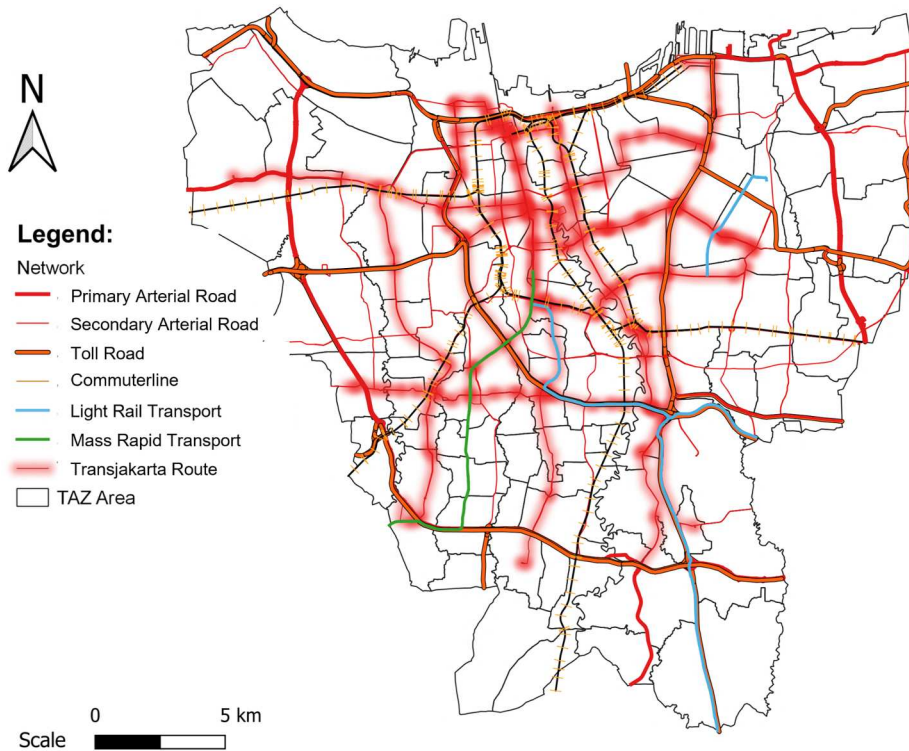


Figure 3. Traffic analysis zones (TAZ) in Jakarta.

the spatial distribution of projected population and other critical socioeconomic indicators within the study area. The TAZ was established through the aggregation of multiple district administrative levels, resulting in a total of 131 TAZ within the study area.

3.2. Zone characteristics

One of the parameters to assess the characteristics of an area is the population and population density (see [Figure 4](#)). The areas with the highest population density are dispersed, specifically on the east, west, and south sides of Jakarta. Regions with high population density on the east side are well-connected by the public transportation network. Population density refers to the number of people residing in an area (population/km²).

3.3. Jobs measurement

The OD matrices were derived from the travel demand forecast model developed using a microsimulation process as part of the JABODETABEK Urban Transportation Policy Integration Project Phase 2 (JICA, 2019). The report obtained a proportion of work-related trips of 65% based on OD interview data, with business trips accounting for 33.9% and work trips for 31.1% (see more details in [Table 1](#)). This proportion is further used as a basis to measure job accessibility for the population from their

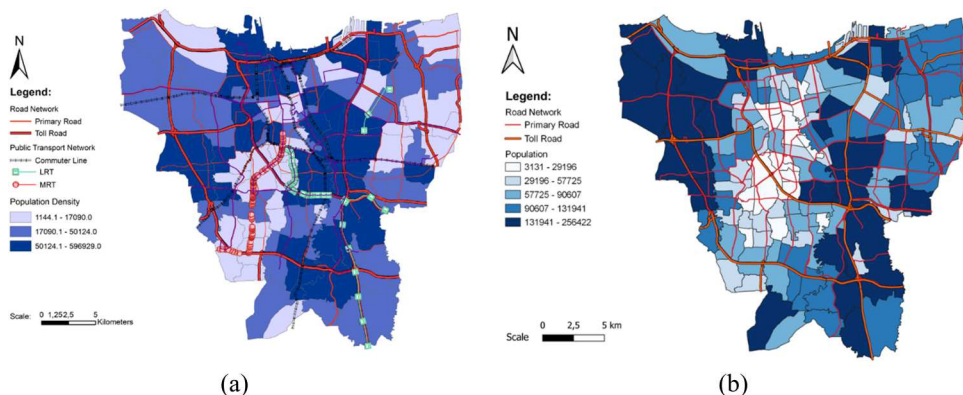


Figure 4. Visual representations of the socioeconomic characteristics in this study: (a) Population Density, (b) Number of Population.

Table 1. Total number of trips and proportions by modes and income levels.

	Car	Motorcycle	Public transport	All modes
High-income	737,695 (5.25%)	1,593,206 (11.34%)	277,343 (1.97%)	2,608,244 (18.56%)
Medium income	2,163,569 (15.40%)	4,813,441 (34.26%)	692,321 (4.93%)	7,669,331 (54.59%)
Low-income	1,064,557 (7.58%)	2,375,155 (16.91%)	332,574 (2.37%)	3,772,286 (26.85%)
All income levels	3,965,820 (28.23%)	8,781,802 (62.50%)	1,302,238 (9.27%)	14,049,860 (100%)

respective origins using OD matrices of private vehicles (cars and motorcycles) and public transport.

According to [Table 1](#), trips are dominated by medium-income households with a proportion of 54.59%, followed by low-income households of 26.85% and high-income households of 18.56%. This distribution of trips highlights the central role of medium-income individuals in shaping travel patterns in Jakarta. Motorcycle trips exhibit a significant level of dominance among all transport modes. Medium-income households, in particular, account for a substantial proportion of motorcycle trips, reflecting their preference for this mode of transportation.

In this study, we assume that the measurement of jobs within a given TAZ can be approximated through the utilization of trip attraction data which reflects the volume of trips directed toward the designated area as their destination. [Figure 5](#) shows the distribution of jobs in Jakarta.

3.4. Accessibility measurement

Infrastructure-based, location-based, person-based, and utility-based accessibility measures are usually used (Geurs & van Wee, 2004). Infrastructure-based measures focus on transport infrastructure and can include supply-oriented metrics like the distance to the next toll gate or demand-and-supply indicators like road network travel times (van Wee et al., 2001). Location-based accessibility estimates cumulative opportunities within a specified travel time or distance or evaluates opportunities at destination locations from an origin, weighted by distance decay. This study employs both infrastructure-based and location-based measures. The infrastructure-based approach assesses the performance of transport infrastructure, focusing on service levels, such as travel times

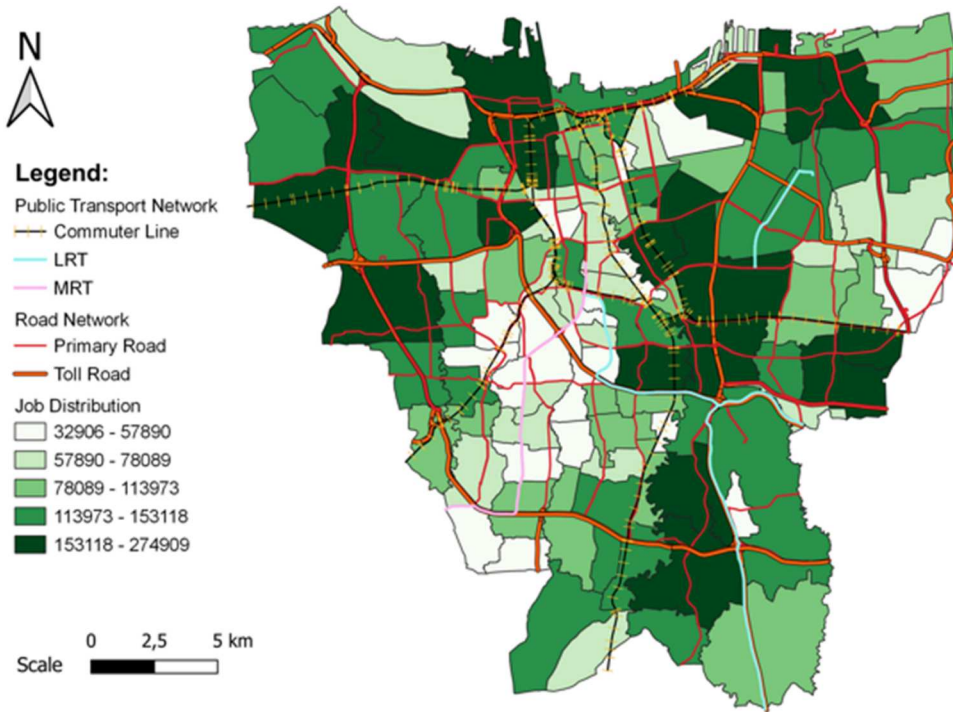


Figure 5. Visual representation of job distribution in Jakarta Province.

on road networks for private vehicles and public transport. Details of the travel time estimation process are provided in the following section.

Job accessibility refers to the ease with which jobs distributed across different locations can be accessed from a specific point. Cumulative job accessibility is calculated using the following formula:

$$A_{i,m,o,s} = \sum_j O_j f(C_{i,j,m}) \quad (1)$$

$A_{i,m,o,s}$ is accessibility in zone i , transport mode m , to activity o in travel time threshold s . O_j is the number of activities in zone j . $C_{i,j,m}$ is the travel time between zone i and j with transport mode m . The impedance function $f(C_{i,j,m})$ is defined as:

$$f(C_{i,j,m}) = \begin{cases} 1, & \text{if } C_{i,j,m} \leq 41.2 \text{ minutes} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

This binary function assigns a value of 1 to destinations reachable within 41.2 min and 0 otherwise. The 41.2-minute threshold reflects the average commuting time in Jakarta, based on data from the Jakarta Urban Mobility survey (2019) and corroborated by Google Maps' mobility statistics. This threshold provides a locally grounded benchmark for evaluating realistic commuting behaviour across different transport modes and socio-economic groups. It ensures consistency in comparing accessibility outcomes while remaining aligned with observed travel patterns in the city.

However, the threshold-based cumulative accessibility approach has certain limitations. First, it imposes a rigid cut-off, which may oversimplify the gradual decline in willingness to travel beyond the threshold. In reality, individuals' tolerance for travel time may vary depending on income, trip purpose, and transport availability. Second, the binary nature of the measure does not account for the qualitative experience of travel – such as reliability, comfort, or safety – factors which may influence accessibility but are not captured in this model. Lastly, the threshold does not consider time-of-day variation or congestion effects, which could lead to under- or over-estimation of actual accessibility during peak hours. Despite these limitations, the threshold method remains a widely used and interpretable measure for highlighting spatial disparities in access and guiding comparative equity assessments.

For the purpose of this research, job accessibility will be calculated for each zone and assessed across transportation modes and income levels, spanning low to high income. Accessibility maps will be categorized into five classes ('very low' to 'very high') using the natural breaks classification in ArcGIS Pro software. Natural breaks classification (Jenks) organizes data into categories based on natural groupings within the dataset. Class breaks are determined to group similar values effectively while maximizing the distinctions between different categories. The categorization method was chosen as it effectively categorizes accessibility into distinct classes based on significant differences in data values, maximizing interpretability with standardized classification for easier comparison across different variables compared to other methods.

3.5. Travel time between zones

This study obtained travel time data for Jakarta's 133 Traffic Analysis Zones (TAZs) using the Google Maps Distance Matrix API, capturing estimates for car, motorcycle, and public transport travel. The data collection process involved the following steps:

(1) Zone Identification and Centroid Mapping

The study utilized 133 TAZs covering the Jakarta Province, with centroid coordinates identified based on spatial aggregation derived from JICA's urban transport planning framework (JICA, 2019). These centroids were used as origin and destination points for travel time estimation.

(2) Google Maps API Configuration and Access

A valid API key from the Google Maps Platform was acquired, enabling access to the Distance Matrix API, which provides travel time estimates incorporating both real-time and historical traffic data.

(3) OD Pair Construction

All possible origin-destination pairs between zones were constructed, yielding 17,689 unique combinations (133×133). For each pair, travel time data were retrieved for three transport modes: car, motorcycle, and public transport.

(4) Parameterization of API Requests

API queries included the following specifications:

- Mode of transport: driving, two_wheeler, or transit
- Departure time: Monday, November 2nd, 2020, at 10:00 am, representing typical mid-morning conditions and avoiding peak-hour distortions.
- Traffic model: Set to 'best_guess' for motorized modes to reflect realistic congestion.

(5) Batch Processing and Rate Limiting

Given the volume of API calls, data collection was automated using Python scripts with looping and batch execution. Rate limiting and error-handling procedures ensured compliance with API quotas and robustness in data retrieval.

(6) Data Extraction and Structuring

API responses in JSON format were parsed to extract travel times, distances, and (for public transport) route information. These data were organized in structured formats (e.g. CSV) for integration into the accessibility analysis.

(7) Integration into Accessibility Model

The processed travel time matrices were used to compute cumulative job accessibility for each mode using a 41.2-minute threshold, reflecting Jakarta's average commute duration. This dataset formed the foundation for the study's spatial, socio-economic, and mobility inequity analyses.

Figure 6 displays the travel time data obtained from the Google Maps API. An individual is travelling from the centroid zone 'Monas' to the nearest centroid zone 'Dukuh Atas 1'. Various transportation modes were considered, including car, motorcycle, public transit, walking, or cycling. Travel time data were specifically collected for car, motorcycle, dan public transit. Public transit assumed that access and egress trips could have been completed by walking due to limitations in the data provided by the Google API. In practice, access and egress trips could have been completed by cycling or via ride-hailing services. The travel time data were retrieved on Monday, November 2nd, 2020, at 10:00 am via Indonesia's Google Maps route search service. This time window reflected typical mid-morning conditions, avoiding peak-hour congestion, and the data were collected prior to any COVID-19-related mobility restrictions. Therefore, the dataset reflected normal, undisturbed traffic conditions and could be considered representative of baseline travel behaviour in urban Jakarta. This approach followed the same methodology and time reference as used in a related study (Safira & Chikaraishi, 2023), ensuring consistency and comparability across findings.

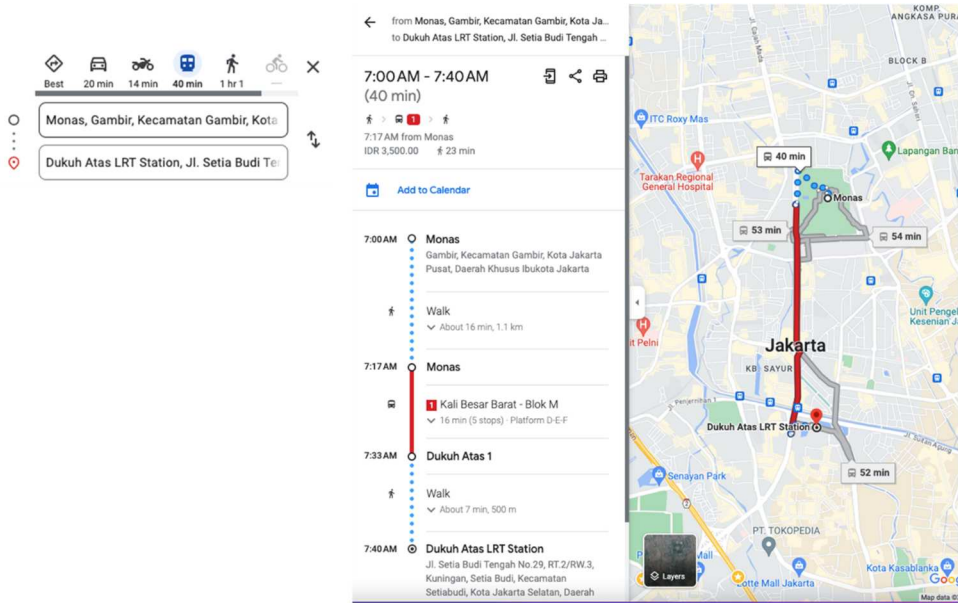


Figure 6. Illustration of travel time data provided by Google API.

3.6. Inequity metrics

3.6.1. Gini coefficient and Lorenz curve

The Gini coefficient measures inequality by comparing the area between the Lorenz curve and the line of perfect equality (a 45-degree line) to the total area beneath it. On a scale of 0 to 1, 0 means everyone has the same share and 1 means all values are concentrated in one person or group.

The Lorenz curve, developed by Lorenz (1905), shows the population's cumulative distribution of a variable like wealth or income. The Gini coefficient quantifies inequality, while the Lorenz curve illustrates it (see Figure 7). Area A between the Lorenz curve and the line of equality is divided by the entire area under the line of equality (A + B) to determine the coefficient. This method permits mathematical inequality comparison across distributions. Lorenz curves and the Gini coefficient can be used on any cumulative distribution, including accessibility levels vs population proportion, as in this case. They are usually employed for income.

The Gini coefficient can be mathematically estimated using the following formula:

$$G_1 = 1 - \sum_{k=1}^n (X_k - X_{k-1})(Y_k + Y_{k-1}) \quad (3)$$

In this context, X_k represents the cumulated proportion of the population variable, where $k = 0, \dots, n$, with $X_0 = 0$, $X_n = 1$. Similarly, Y_k denotes the cumulated proportion of the access levels, for $k = 0, \dots, n$, with $Y_0 = 0$, $Y_n = 1$ (Delbosc & Currie, 2011).

3.6.2. Palma ratio

Palma (2011) introduced the Palma ratio to compare the income share of the richest 10% to that of the poorest 40%. However, as zone-level income data is unavailable in

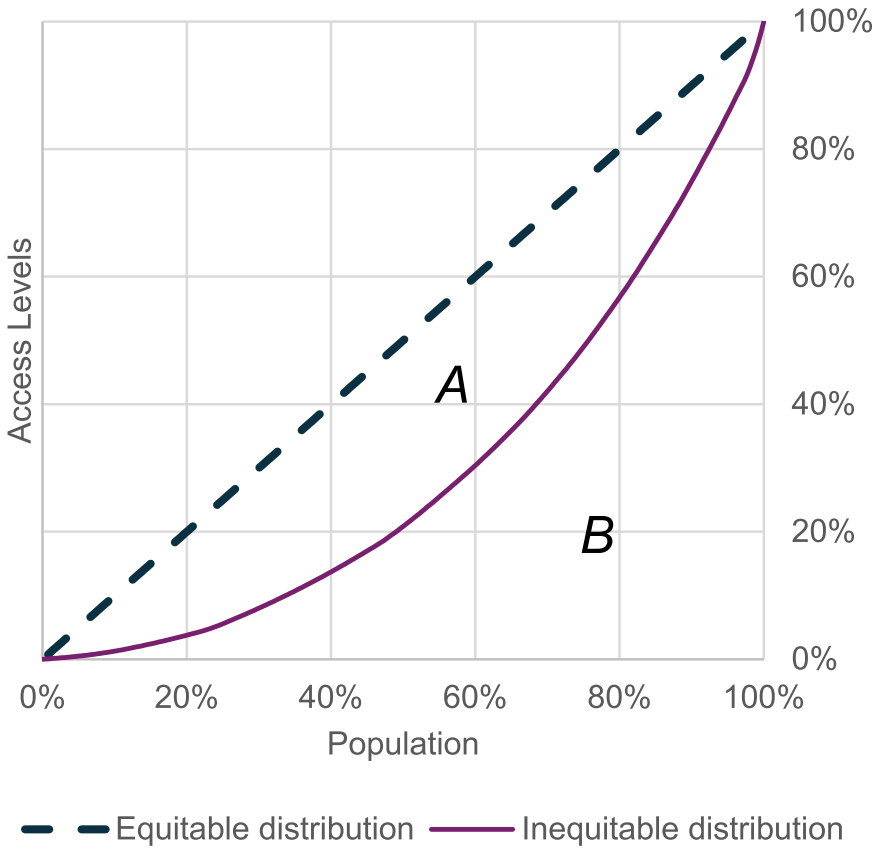


Figure 7. Lorenz curve example.

this context, the ratio is adapted to evaluate disparities in job accessibility. Specifically, this study calculates the ratio using the average accessibility of the 10% most densely populated TAZs and the 40% least densely populated TAZs. This approach shifts focus away from middle-density groups, emphasizing disparities between the extremes.

$$P = \frac{\bar{A}_{i_{top10}}}{\bar{A}_{i_{bottom40}}} \quad (4)$$

$\bar{A}_{i_{top10}}$ is the average accessibility of the areas with the highest 10% population density,

$\bar{A}_{i_{bottom40}}$ is the average accessibility of the areas with the lowest 40% population density

A Palma ratio of 2.0 suggests that the top 10% of areas, ranked by population density, have job access levels that are double those of the bottom 40%.

3.6.3. Theil index

This study utilized the Theil index to assess inequity in job accessibility. The Theil index is particularly effective for this analysis because it allows the inequity to be broken down into components representing disparities within and between different population groups. The study adopts a stratification based on three population density groups in

Jakarta: low density (< 17,090 population/km²), medium density (17,090 - 50,123 population/km²), and high density ($\geq 50,124$ population/km²).

The overall Theil index T , along with its within-group (T_{WG}) and between-group (T_{BG}) components can be determined using the following formula:

$$T = \frac{1}{N} \sum_i^N \frac{A_i}{A} \ln \left(\frac{A_i}{A} \right) = T_{BG} + T_{WG} \quad (5)$$

$$T_{BG} = \sum_{k=1}^K A_k \ln \left(\frac{A_k}{\frac{n_k}{N}} \right) \quad (6)$$

$$T_{WG} = \sum_{k=1}^K A_k \left[\sum_{i \in g_k} \frac{A_{ki}}{A} \ln \left(\frac{\frac{A_{ki}}{A_k}}{\frac{1}{n_k}} \right) \right] \quad (7)$$

In this context, N represents the total number of grids; A_i is the accessibility value of grid i ; A denotes the average accessibility of all the grids. The parameter K indicates the number of groups, with g_k representing the set of grids whose centroids fall within the group k , and n_k being the total number of grids in g_k . In addition, A_{ki} and A_k correspond to the proportion of the total accessibility for grid i and group k respectively (Jin et al., 2022).

4. Results and discussion

4.1. Patterns of job accessibility in Jakarta

Job accessibility will be quantified for each Traffic Analysis Zone (TAZ) by determining the number of jobs reachable within a specified time travel threshold by the population residing in the corresponding TAZ. The threshold used for this analysis is determined by averaging the time travel data across all transport modes (car, motorcycle, and public transport), resulting in a threshold of 41.2 min. Utilizing the number of jobs available in a TAZ by trip attraction data, as discussed in section 3.4, we further calculate the number of jobs that each TAZ, serving as an origin, can access within the established 41.2-minute time frame. Figure 8 shows the compilation of maps providing a general comparison of job access levels across various means of travel and income levels.

4.1.1. Spatial distribution of job accessibility

The spatial distribution of job accessibility in Jakarta reveals pronounced centrality, with the highest accessibility concentrated in the city's urban core, particularly in Central, South, and parts of West Jakarta. These areas host Jakarta's primary employment clusters, commercial centres, and administrative districts, making them critical focal points for commuting flows. Accessibility consistently declines toward the northern, eastern, and peripheral southern areas, especially in zones where infrastructure is inadequate or disconnected from public transport networks.

Among all transport modes, motorcycles offer the widest spatial coverage, enabling access to a broad range of jobs from both central and peripheral areas. This spatial

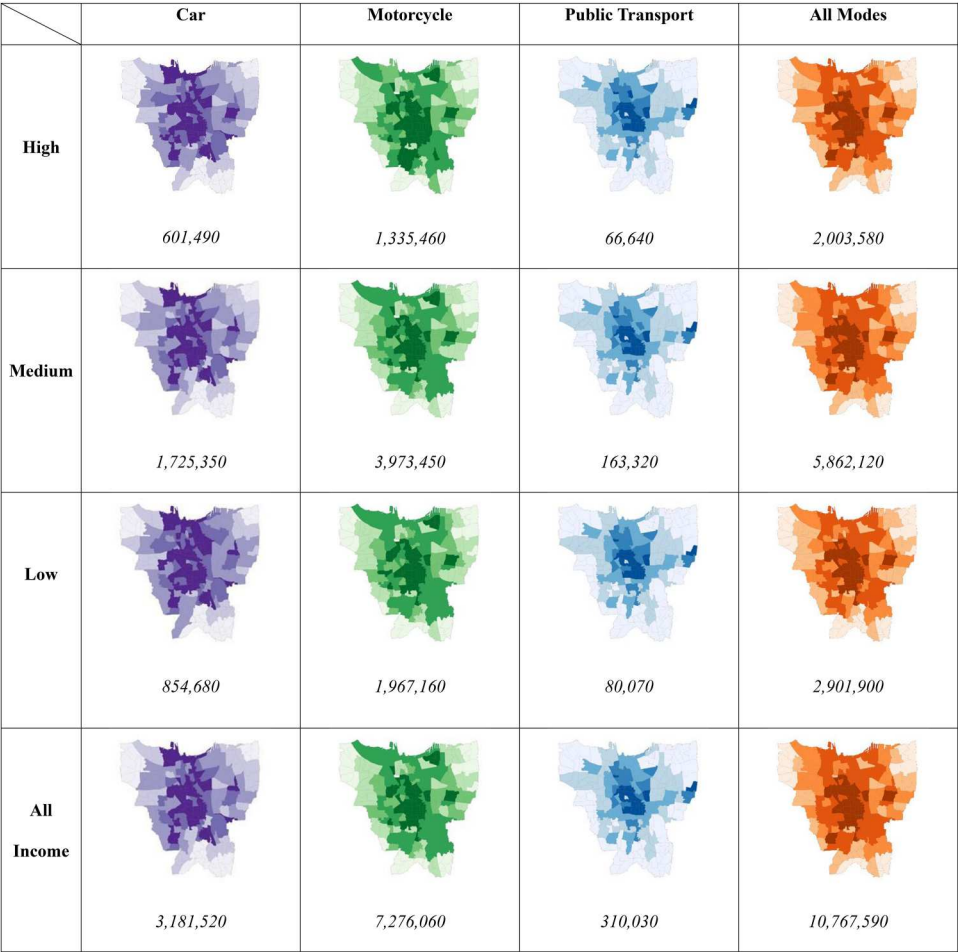


Figure 8. Visual comparison and average values of job access levels by transport modes and income levels in Jakarta.

flexibility allows motorcycles to reach job centres not only in well-developed districts but also in semi-urban or informal settlements where public transport coverage is weak. This pattern reflects broader trends in Global South cities, where informal or two-wheeler modes bridge spatial gaps in formal transport systems (Uteng & Lucas, 2017).

Conversely, public transport accessibility is spatially limited, primarily concentrated along a few high-capacity transit corridors served by bus rapid transit (BRT) and commuter rail lines. While accessibility is relatively high near these corridors, it quickly declines in areas located beyond the limited reach of trunk lines and transfer points. The absence of a comprehensive feeder system and inadequate first- and last-mile infrastructure contribute to the spatially selective nature of public transport access, reinforcing exclusion for residents in underserved peripheral districts.

Car accessibility exhibits spatial selectivity, concentrated in areas with high-quality road infrastructure and less congestion. However, Jakarta’s chronic traffic delays and

poor road network in dense informal settlements constrain car-based accessibility, particularly in lower-income neighbourhoods.

4.1.2. Modal contributions to job accessibility

The mode of transport is a critical determinant of job accessibility outcomes. Motorcycles emerge as the dominant enabler, accounting for 62.5% of all jobs reached in the region. Their role is particularly important for medium- and low-income populations, who together rely on motorcycles to access over 7 million jobs. The efficiency, affordability, and adaptability of motorcycles in Jakarta's congested and often fragmented urban fabric make them a cornerstone of urban mobility (Joewono & Kubota, 2007).

This study demonstrates that motorcycles markedly improve job accessibility for all socio-economic groups, supporting the findings of Irawan et al. (2020), which indicate that motorcycle ride-hailing services enhance urban mobility for lower-income households. However, this equity comes at a cost. Motorcycles are also associated with high emissions, road safety risks, and unregulated traffic behaviours, raising critical questions about the sustainability of equity gains achieved through informal transport (Gutierrez & Mohan, 2020; Kalifa et al., 2024; Mahesh et al., 2019; Yousif et al., 2020). This duality shows the complex trade-offs facing urban transport policy in rapidly motorizing cities – where motorcycles function both as an equalizer and a sustainability challenge.

The heavy reliance on motorcycles also reflects systemic deficiencies in the formal transport system, particularly in delivering broad, inclusive access to employment. Motorcycles fill the vacuum left by inadequate public transport coverage, especially in areas beyond the reach of trunk routes.

In contrast, public transport plays a minimal role, facilitating access to only 9.27% of total jobs. Even for the most advantaged group (medium-income), public transport allows access to just 692,000 jobs (4.93%). The system's limited geographic coverage, weak integration, and low service frequency constrain its potential to equitably connect residents with economic opportunities (as also argued by Basso et al., 2020; Bueno Rezendede Castro et al., 2022). These issues disproportionately affect low-income residents who are most dependent on affordable public transit options.

Car-based accessibility, while more substantial than public transport in absolute numbers, also reveals inequality. Medium-income users benefit the most (2.16 million jobs), but high-income users access significantly fewer jobs (737,000) despite greater ownership rates (Sefriyadi et al., 2023). This counterintuitive pattern likely stems from spatial mismatch between elite residential enclaves and job centres, along with the limitations of Jakarta's car-oriented infrastructure in accommodating efficient, long-distance commuting (Oviedo & Guzman, 2020; Saraiva & Barros, 2022).

4.1.3. Income-level differences in job accessibility

Income level is closely intertwined with both spatial and modal patterns of job accessibility. Medium-income individuals consistently demonstrate the highest levels of job accessibility, reaching over 7.67 million jobs (54.59%). This outcome is attributable to both spatial self-selection and greater modal flexibility. Unlike high-income groups, medium-income individuals are often willing to reside in modest housing conditions closer to employment centres, striking a balance between affordability and accessibility (Adeniran et al., 2024; Friedman et al., 2020; Levine, 1998). Their neighbourhoods are

frequently located in older, mid-density areas that remain well-connected to road and transport networks.

High-income residents, although financially advantaged, access fewer jobs (18.56%) due to residing in low-density, exclusive zones, often on the periphery or in gated communities. Aritenang (2022) observed that residents with higher incomes, despite living in peripheral zones, often commute to central urban areas. These locations prioritize privacy and housing quality over proximity to work and are often disconnected from both public and informal transport routes. As a result, their effective job accessibility remains constrained.

Low-income residents – while potentially located closer to the city core – face barriers of a different kind. Limited financial capacity, inadequate infrastructure, and dependence on low-cost, informal transport restrict their job access to 3.77 million jobs (26.85%). The minimal contribution of public transport for this group further reinforces structural disadvantages, as they are priced out of formal alternatives and often face longer travel times, safety issues, and regulatory uncertainties (Bergman et al., 2024).

4.2. Distributional assessment using inequity metrics

While the previous section detailed variations in job accessibility by mode, income group, and location, such descriptive analysis alone is insufficient to understand the deeper patterns of transport-related disadvantage. Not all differences in accessibility are inherently unfair; therefore, to assess whether these disparities reflect systemic inequities, it is necessary to apply formal distributional metrics. Inequity indicators offer a quantitative lens to measure the extent to which job accessibility is unevenly distributed across different population groups and spatial units, and to identify whether these differences are disproportionately concentrated among certain social or geographic categories.

The measurement of inequity in job accessibility is based on three indices: the Gini coefficient, the Palma ratio, and the Theil index. Analysis of these metrics reveals that access to employment via public transport demonstrates the greatest degree of inequality relative to other transportation modes.

The Gini coefficient and Palma ratio are calculated for each income level, allowing for an analysis of accessibility distribution across different socioeconomic strata. The metrics highlight ongoing challenges in attaining equitable job access, with public transport demonstrating the greatest level of inequality across all income brackets. Furthermore, the Theil index utilizes population density groupings to examine access disparities, distinguishing between within-group and between-group inequalities.

4.2.1. Gini coefficient and Lorenz curve

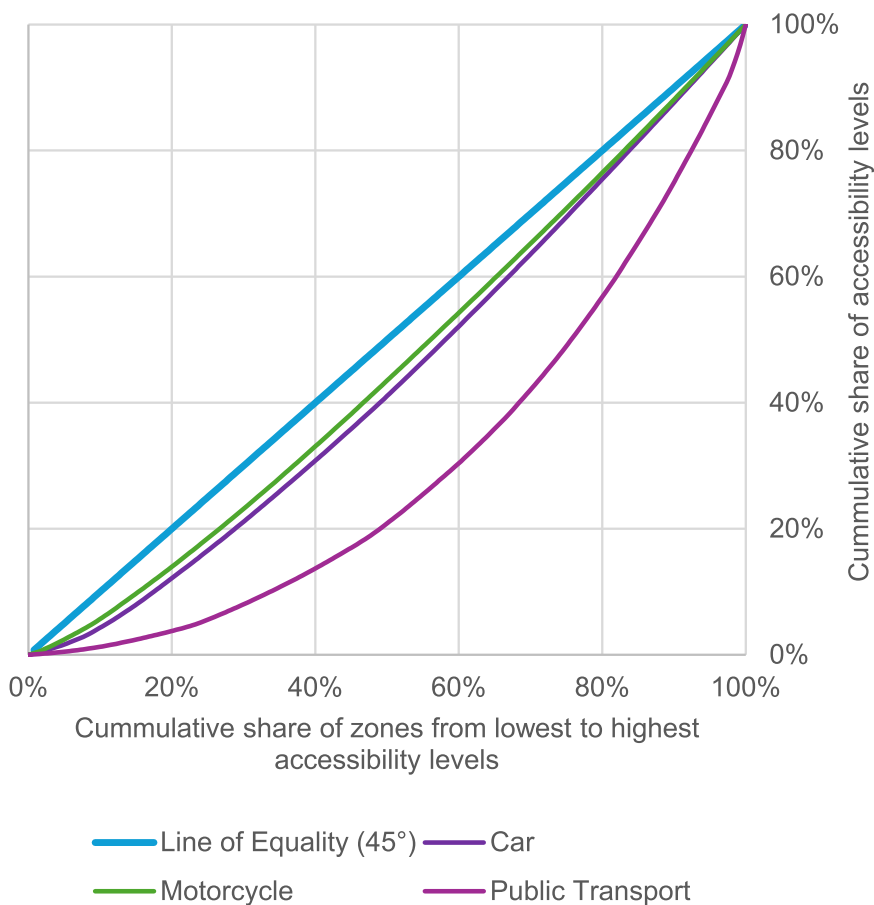
As shown in Table 2, job accessibility by public transport exhibits the highest Gini coefficient (0.417), in contrast to lower values for car (0.140) and motorcycle (0.109). These results highlight that access via public transport is the most unevenly distributed. Figure 9 further visualizes this disparity; the Lorenz curve for public transport deviates furthest from the line of equality, indicating that a large share of job access is concentrated in a small subset of zones.

These findings align with research from Global South cities, where public transport networks often serve dense, high-income corridors while peripheral or informal

Table 2. Gini coefficients of job accessibility distribution.

Income level	Car	Motorcycle	PT	All modes
High-income	0.137	0.108	0.424	0.116
Medium income	0.142	0.110	0.414	0.117
Low-income	0.140	0.109	0.421	0.117
All income levels	0.140	0.109	0.417	0.117

Lorenz Curve of Job Accessibility Distribution

**Figure 9.** Lorenz curve of job accessibility distribution by transport modes.

settlements remain underserved, exacerbating spatial inequity (Boisjoly et al., 2020; Gumbo & Moyo, 2020). In contrast, the motorcycle curve lies much closer to the line of equality, consistent with its lower Gini coefficient and reinforcing its role in bridging formal service gaps due to affordability and geographic reach (Carrales, 2024; Irawan et al., 2020).

Interestingly, the Gini coefficients remain remarkably consistent across income groups for each mode of transport. For instance, public transport access yields a Gini of 0.424 for high-income, 0.414 for medium-income, and 0.421 for low-income users.

This indicates that there are no significant differences in the distribution of job accessibility levels across high-, medium-, and low-income individuals, suggesting that the structure of inequity is similar across socio-economic strata. A comparable trend is observed for motorcycles and cars, where the values vary only slightly across income levels.

These results imply that inequity is not inherently tied to income class, but rather to mode-specific characteristics and spatial transport infrastructure. Each income group exhibits internal stratification, where some zones benefit substantially more than others. As such, transport equity policies must go beyond income targeting and also address intra-group, neighbourhood-level disparities that exist even within the same socio-economic classifications.

4.2.2. Palma ratio

The Palma ratio provides additional insight into distributional extremes by comparing the share of job accessibility held by the top 10% of zones to that of the bottom 40%. As shown in Table 3, the Palma ratios for car (1.610), motorcycle (1.442), and public transport (7.361) all exceed 1.0, indicating that accessibility is concentrated among the most advantaged zones across all modes. However, the disparity is most severe for public transport, where the top 10% of zones capture more than seven times the accessibility of the bottom 40%.

This finding is consistent with spatial mismatch theory (Andersson et al., 2011), which posits that low-income households in peripheral or low-access zones are spatially disconnected from employment opportunities, especially when high-quality public transport is concentrated in central districts. In the case of Jakarta, the spatial mismatch is not necessarily due to cost barriers, as observed in cities like Bogota (Guzman & Oviedo, 2018), but rather due to limited network coverage and inadequate integration between transit modes. While public transport may be nominally affordable, its geographic inaccessibility renders it functionally exclusionary for large segments of the population.

Moreover, these results align with those from the Gini coefficient analysis, reinforcing that public transport is the most unequal mode in terms of accessibility distribution. In contrast, motorcycle accessibility is the most evenly distributed, with a relatively low Palma ratio and Gini coefficient. This again emphasizes the unique role motorcycles play in expanding access in areas poorly served by formal transport modes.

Similar with the Gini results, the Palma ratios for all three income groups (low, medium, and high) show minimal variation, suggesting that inequities in accessibility are not strongly differentiated by income strata. Rather, they reflect a broader pattern of modal and spatial exclusion, confirming earlier observations that transport inequity in Jakarta is shaped more by geography and network design than by socio-economic category alone.

Table 3. Palma ratio of job accessibility distribution.

Income level	Car	Motorcycle	PT	All modes
High-income	1.584	1.429	7.616	1.501
Medium income	1.622	1.448	7.233	1.513
Low-income	1.609	1.442	7.486	1.505
All income levels	1.610	1.442	7.361	1.508

Table 4. Theil index of job accessibility distribution.

Theil	Car	Motorcycle	PT
Within Group (TW)	0.032 (91%)	0.018 (92%)	0.265 (96%)
Between Group (TB)	0.003 (9%)	0.002 (8%)	0.010 (4%)
Theil Index (TB + TW)	0.036	0.019	0.275

4.2.3. Theil index: within-group vs. between-group inequity

The Theil index adds another layer of insight by decomposing inequity into within-group and between-group disparities. As shown in Table 4 and Figure 10, 96% of total inequality in public transport access arises within population density groups, compared to only 4% between them. Similar patterns are observed for cars (91% within-group) and motorcycles (92% within-group).

This shows that inequality cannot solely be caused by variations between high- and low-density locations; it’s also caused by variances within each group. This means that some neighbourhoods in dense areas remain without access to jobs because of inadequate connectivity or gaps in infrastructure. Thus, broad spatial categories like ‘central vs. peripheral’ or ‘dense vs. sparse’ may obscure crucial differences that are only found in certain areas. City-wide or corridor-scale changes alone are unlikely to render mobility more fair. Fine-grained improvements at the district or subdistrict level, including feeder routes, pedestrian access to stations, or micro-mobility infrastructure, may work better to reduce spatial exclusion.

These results go against the popular belief in urban planning that density alone assures access and show how important it is to make changes at the neighbourhood level. Policies for public transit in cities frequently focus on expanding the network. However, this study implies that making access easier and consolidating it in places that are already ‘served’ is just as important. This entails making stations work better together,

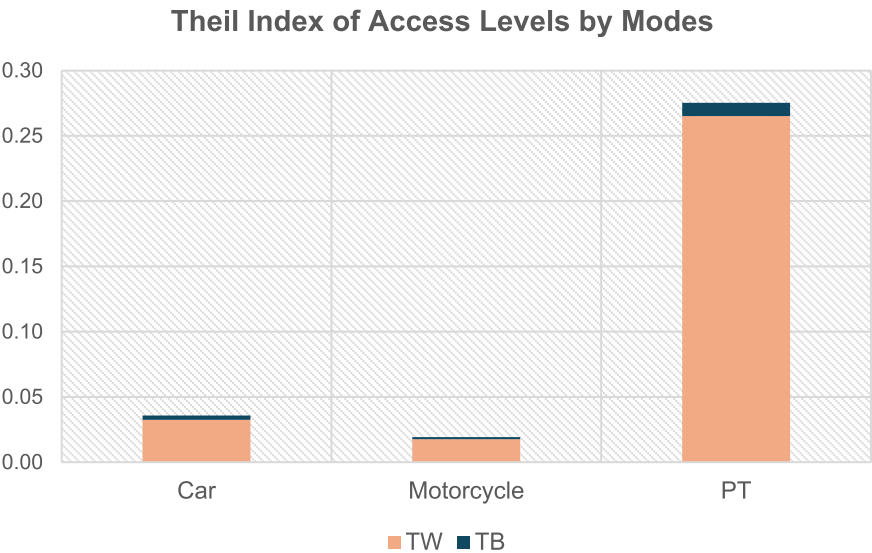


Figure 10. Theil index of job accessibility distribution by transport modes.

running more often, and connecting different modes of transportation in areas where public transportation is available but fails to deliver equitable outcomes.

5. Conclusion

This study offers a comprehensive, multi-dimensional assessment of job accessibility in Jakarta, analysing spatial, modal, and income-related disparities through the use of a cumulative opportunities framework and equity metrics, including the Gini coefficient, Palma ratio, and Theil index. The results indicate a significant spatial centrality in job accessibility, with residents of Jakarta's urban core having considerably more access to employment opportunities than those in peripheral and low-density areas and remaining underserved. Transportation mode plays a critical role in this divide.

Motorcycles offer the most comprehensive and equitable employment accessibility among all modes. They are a critical means of access for medium- and low-income groups due to their flexibility and reach into both central and peripheral zones. Nevertheless, this dependence on motorcycles also underscores the systemic deficiencies in Jakarta's formal transportation infrastructure and raises long-term concerns regarding safety, sustainability, and regulation.

Conversely, public transportation exhibits the most severe inequality. Accessibility via public transport is spatially concentrated in a small number of well-served zones and does not extend effectively to underserved areas. This is evidenced by its high Gini coefficient (0.417), extreme Palma ratio (7.361), and Theil index, which suggests that over 90% of inequality is concentrated within population density groupings rather than between them. These findings suggest that spatial exclusion in Jakarta is not only a result of large-scale urban structure but also of neighbourhood-level service deficiencies, reinforcing the need for localized, fine-grained improvements such as feeder services, pedestrian access, and micro-mobility infrastructure.

While medium-income individuals enjoy the highest average access, the similarity in Gini coefficients across income categories indicates that income alone is not the dominant factor shaping inequity. Instead, geographic location and transport mode are more influential, pointing to spatial and modal exclusion as the primary drivers of job accessibility disparities in Jakarta.

Beyond its empirical contributions, this study makes three important additions to the transport equity literature. First, it presents a replicable framework for evaluating job accessibility in the Global South using formal, multidimensional equity metrics, offering one of the first such applications in a Southeast Asian megacity. Second, it integrates spatial, socio-economic, and modal dimensions into a unified analytical approach, enabling a more comprehensive understanding of transport-related disadvantage. Third, it draws attention to context-specific mobility patterns – such as motorcycle dependence and fragmented transit networks – that are often overlooked in global accessibility research. In doing so, the study responds directly to the need, articulated in the literature, for context-sensitive and data-driven assessments of transport inequality in emerging urban regions.

The findings highlight the urgency of expanding an inclusive and integrated public transport network, improving first- and last-mile connections, and extending service coverage to underserved districts. While motorcycles may continue to serve as an

adaptive mobility solution, long-term transport planning should prioritize the transition toward safer, low-emission, and more reliable public modes. Embedding equity metrics into transport planning and investment decisions is essential to identify and address intra-urban disparities and ensure that accessibility gains are equitably shared.

Future research should explore how job accessibility and transport equity evolve over time, especially in response to Jakarta's ongoing investments in MRT, LRT, and BRT systems. Longitudinal studies comparing pre- and post-intervention conditions would provide evidence on whether infrastructure investments effectively reduce inequity. Further research should also incorporate more granular socio-demographic variables – such as gender, age, employment type, and disability status – to examine how different groups experience transport exclusion. Methodologically, the development of dynamic accessibility models that reflect real-time service variability could enhance the realism of future assessments. Lastly, future work should explore the intersection of housing affordability, land use, and transport accessibility, particularly to assess the equity impacts of urban regeneration and transit-oriented development strategies in Jakarta.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Quillbot, developed by Learneo, Inc., to assist with paraphrasing and refining the language in certain sections. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the final content of the publication.

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